MACHINE LEARNING IN PHYSICS

FOUNDATIONS 1

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PHY6937 / PHY4936

Goals of this Course

1. Gain a good understanding of the mathematical basis of machine learning (ML), which is a foundational component of artificial intelligence (AI).

2. Gain experience building ML models using **PyTorch** to solve data science problems in physics.

3. Gain experience with different ML models.

4. Gain an understanding of AI agents.

Course Evaluation

In addition to your Course Participation (attending, asking questions, contributing to discussions), your course grade will be based on:

1. In-class Assignments

2. Project with a write-up and a brief presentation
    1. Graduates: A project related to your research area.
    2. Undergraduates: A project selected from a few options.
    3. Presentations: Last week of the semester.

3. Grade: Graduates (S/U), Undergraduates (Letter Grade).

What is Artificial Intelligence?

Artificial Intelligence

Algorithms that cause machines to exhibit *human*- or *superhuman*-level intelligence.

ChatGPT



Developer(s)	OpenAI
Initial release	November 30, 2022
Stable release	August 7, 2025
Engine	GPT-5

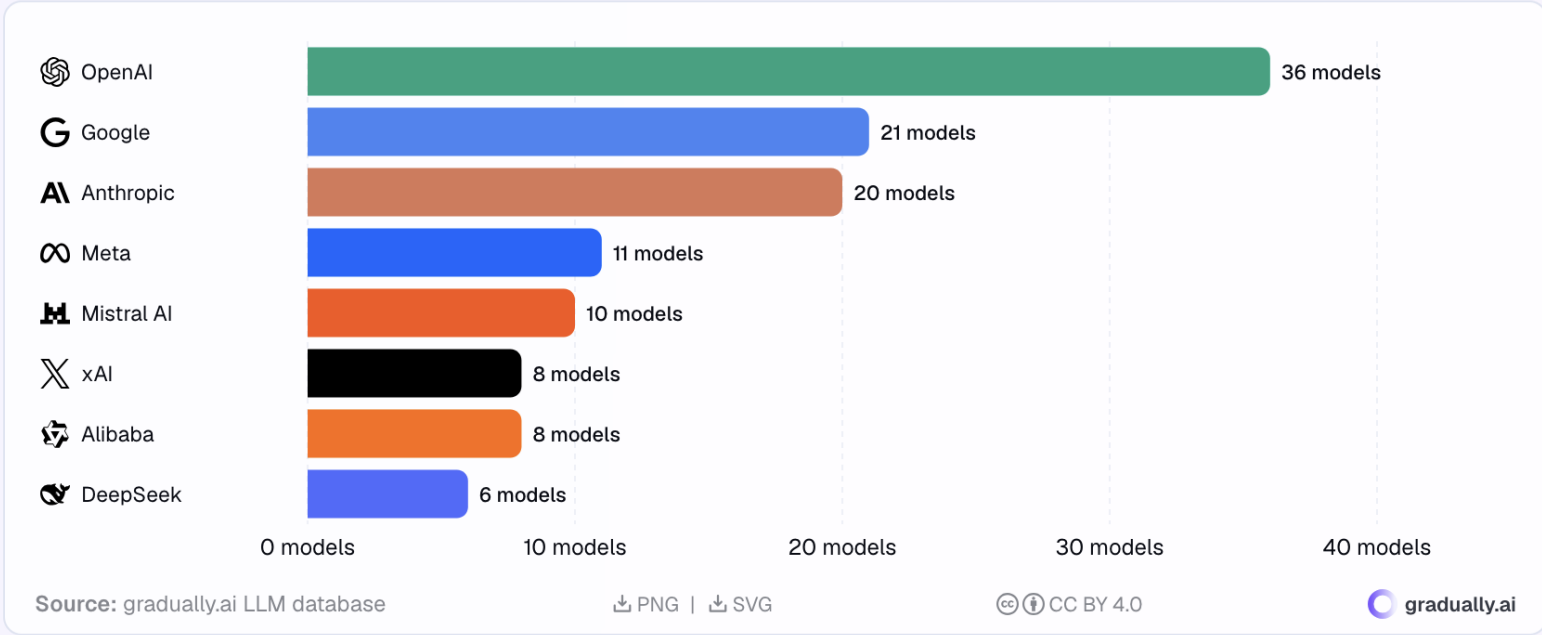
Large Language Model (LLM)



Large Language Model (LLM)

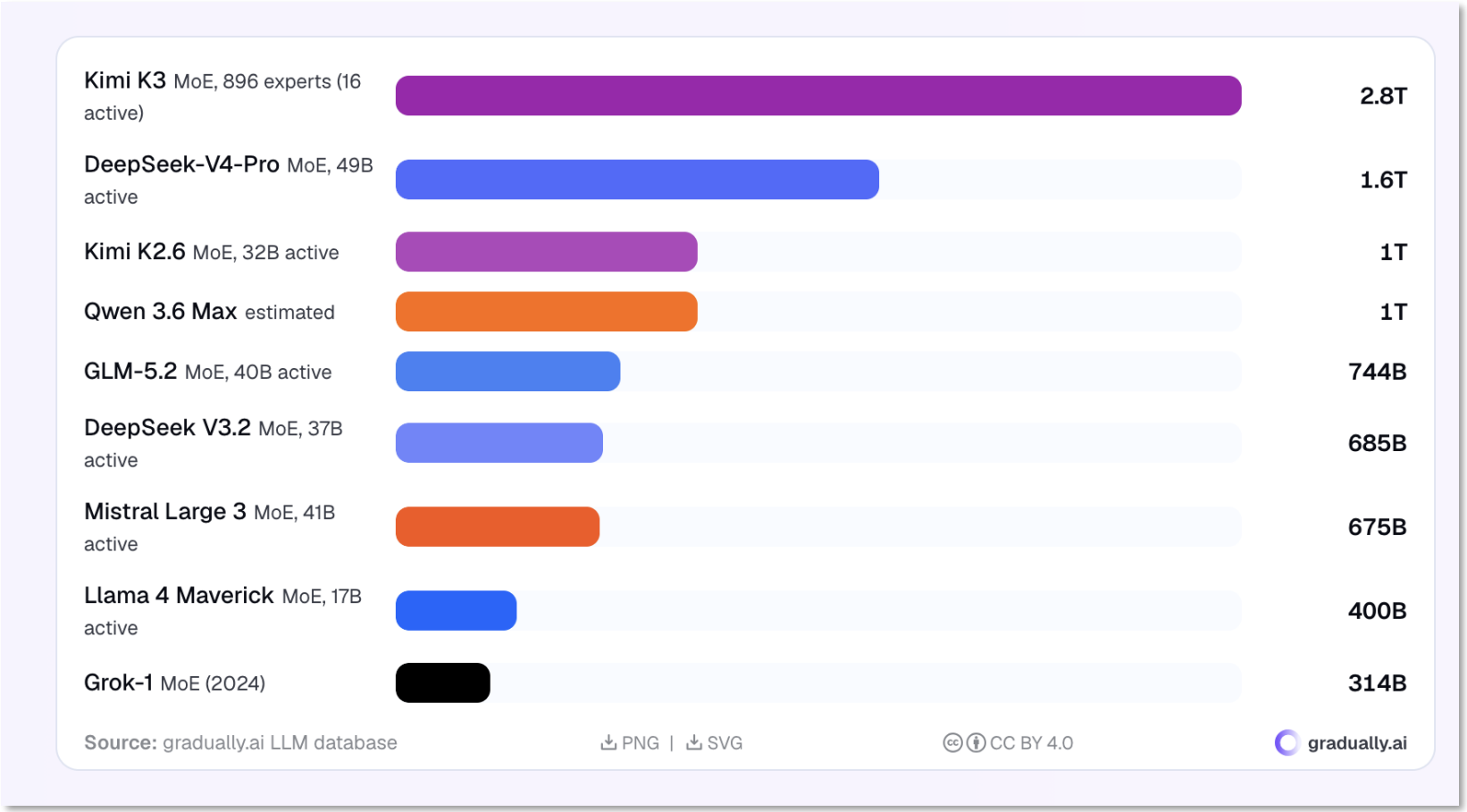
2. The Biggest LLM Providers by Model Count

A simple indicator of how active a lab is: the number of models it maintains. The chart below shows how many of the models we track belong to each provider:



https://www.gradually.ai/en/llm-statistics/

Model sizes: Number of parameters



https://www.gradually.ai/en/llm-statistics/

Claude Demo

What is Machine Learning?

Artificial Intelligence
Algorithms that cause machines to exhibit human- or *super-human* level intelligence.

Machine Learning
Algorithms for modeling data.

What is Machine Learning?

Supervised Learning

Data: (x, y)

y are labels

Task: $x \rightarrow y$

Use cases:

- Classification, regression, translation, etc.

Unsupervised Learning

Data: x

no labels

Task: find structure in, and/or model, data

Use cases:

- Clustering, data compression, solving differential equations, etc.

What is Machine Learning?

Generative Learning

Data: x

may or may not be associated with labels

Task: $x \rightarrow p(x) \rightarrow x$

Use cases:

- fast simulators, image/text generation, chatbots, etc.

Reinforcement Learning

Data: (x, a, r)

x state of the environment

a action taken on environment

r reward arising from action

Task: find optimal $x \rightarrow a$

Use cases:

- Game playing, robotics, accelerator controls, etc.

What is Deep Learning?

Artificial Intelligence
Algorithms that cause machines to exhibit human- or *super-human*-level intelligence.

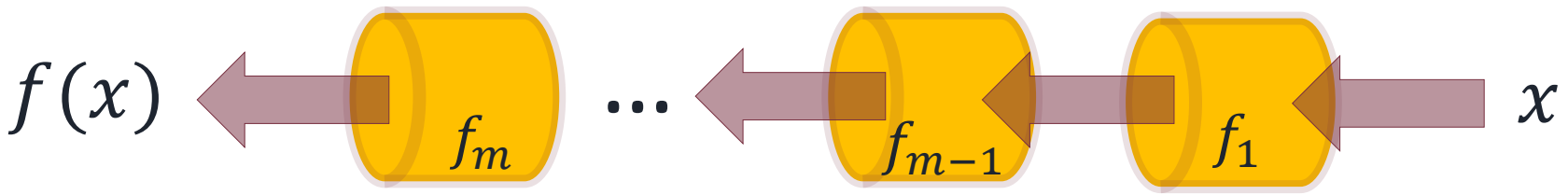
Machine Learning
Algorithms for modeling data

Deep Learning
ML using (large) neural networks

What is Deep Learning?

Deep learning is the science and art of fitting models to data using functions formed through *composition*:

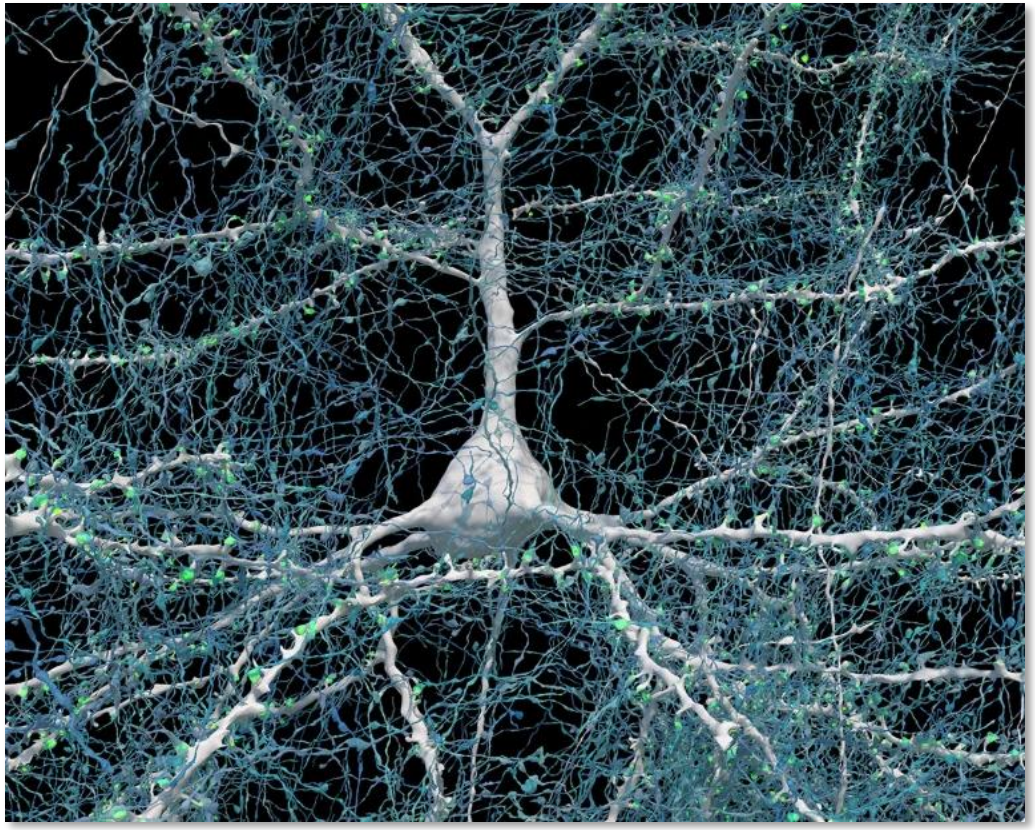
$$f(x) = f_m \circ f_{m-1} \circ \cdots f_1$$
$$= f_m(f_{m-1}(\ldots f_1(x)) \ldots)$$



Each of these functions is referred to as a **layer**. Kimi K3, from Moonshot AI, has **93** layers and **2.8 trillion** parameters!

BASIC BUILDING BLOCK:
THE PERCEPTRON

The Brain’s Computational Unit



https://www.nature.com/articles/d41586-024-01387-9

Basic AI Comp. Unit: The Perceptron

$$y = g(xA^T + b)$$

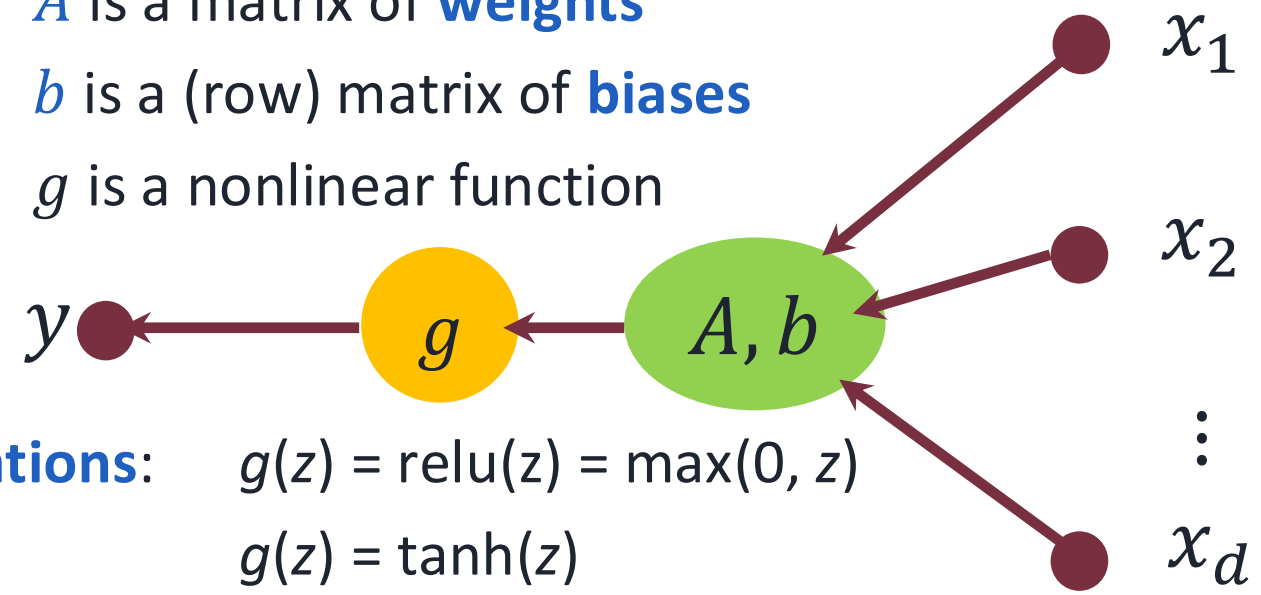
(Frank Rosenblatt, 1958)

x is a (row) matrix of input data

A is a matrix of **weights**

b is a (row) matrix of **biases**

g is a nonlinear function



Activations: $g(z)$ = relu(z) = max(0, z)

$g(z)$ = tanh(z)

$g(z)$ = sigmoid(z) = 1 / (1 + exp(-z))

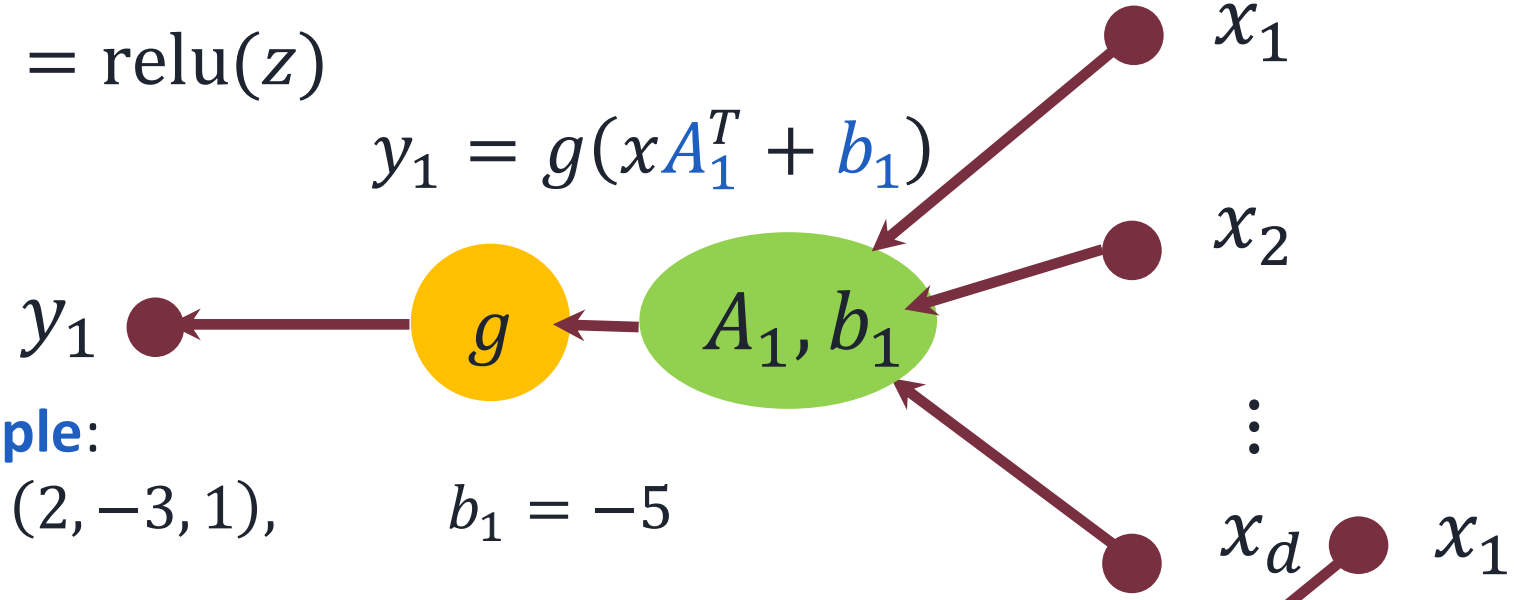
https://news.cornell.edu/stories/2019/09/professors-perceptron-paved-way-ai-60-years-too-soon

MULTI-NODE PERCEPTRON

Multi-Node Perceptron

$g(z) = \text{relu}(z)$

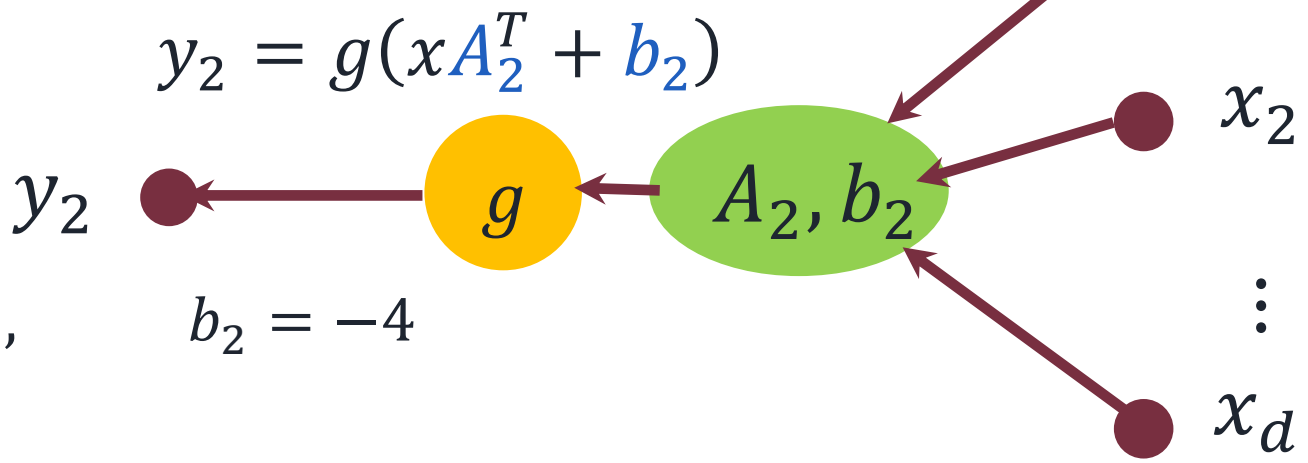
$y_1 = g(xA_1^T + b_1)$



Example:

$A_1 = (2, -3, 1),$ $b_1 = -5$

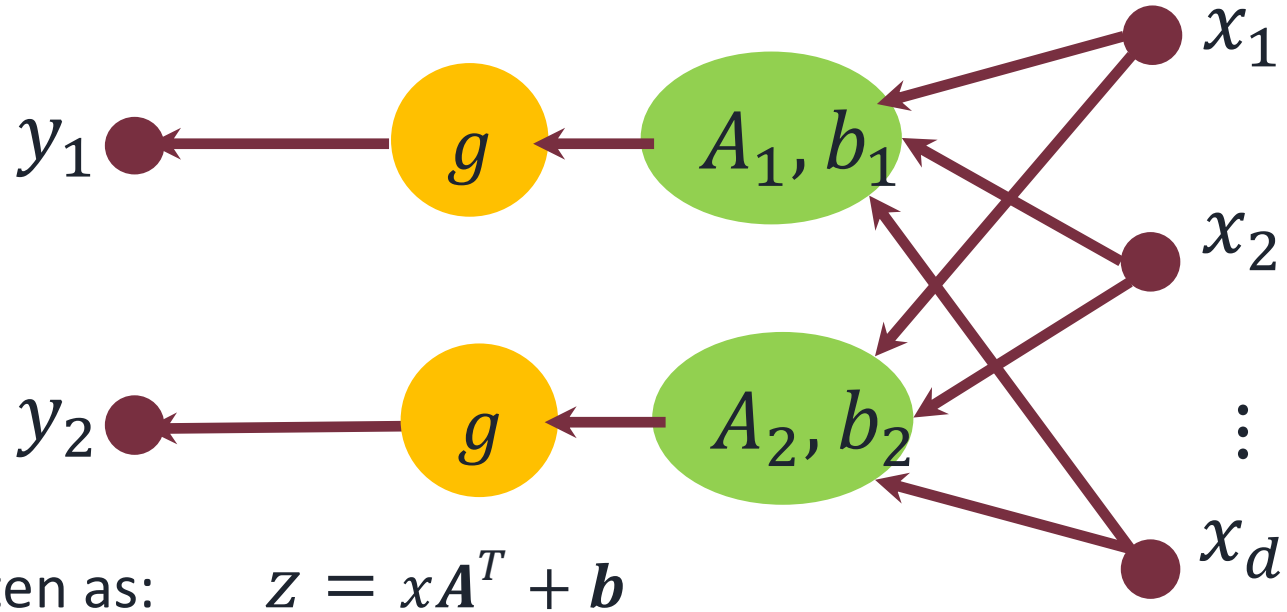
$y_2 = g(xA_2^T + b_2)$



$A_2 = (1, 2, 3),$ $b_2 = -4$

Multi-Node Perceptron

A multi-node perceptron is often drawn as follows,

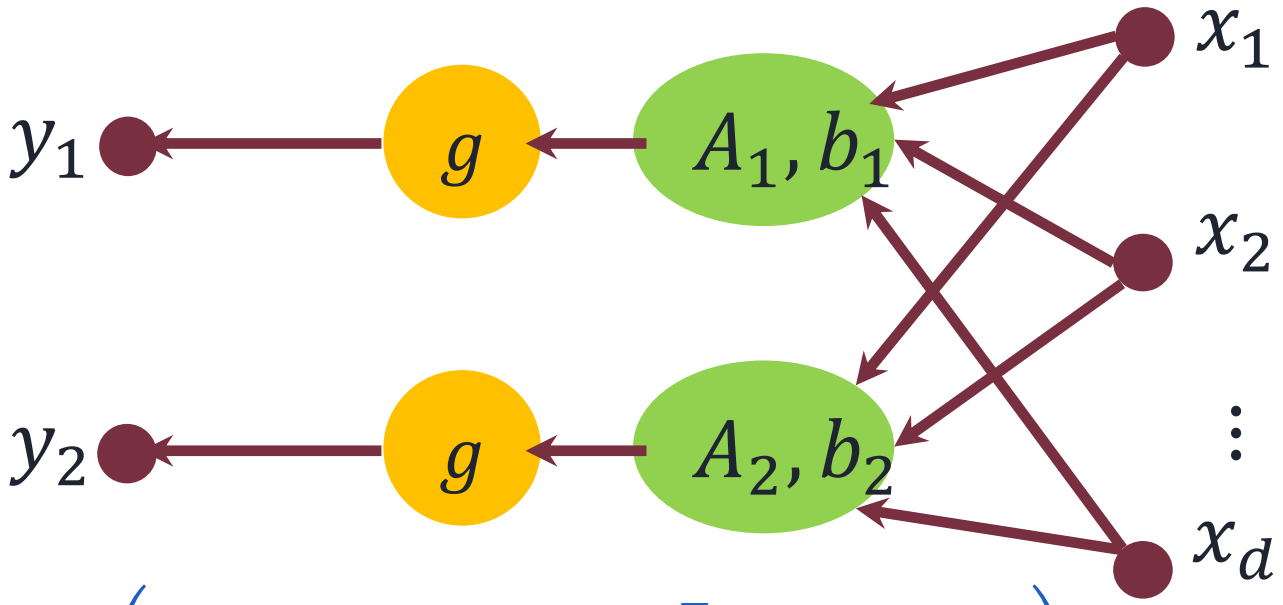


Written as: $z = xA^T + b$

with $y = g(z)$, usually, the applied *elementwise*, that is, to every element of its matrix input.

Multi-Node Perceptron: Example

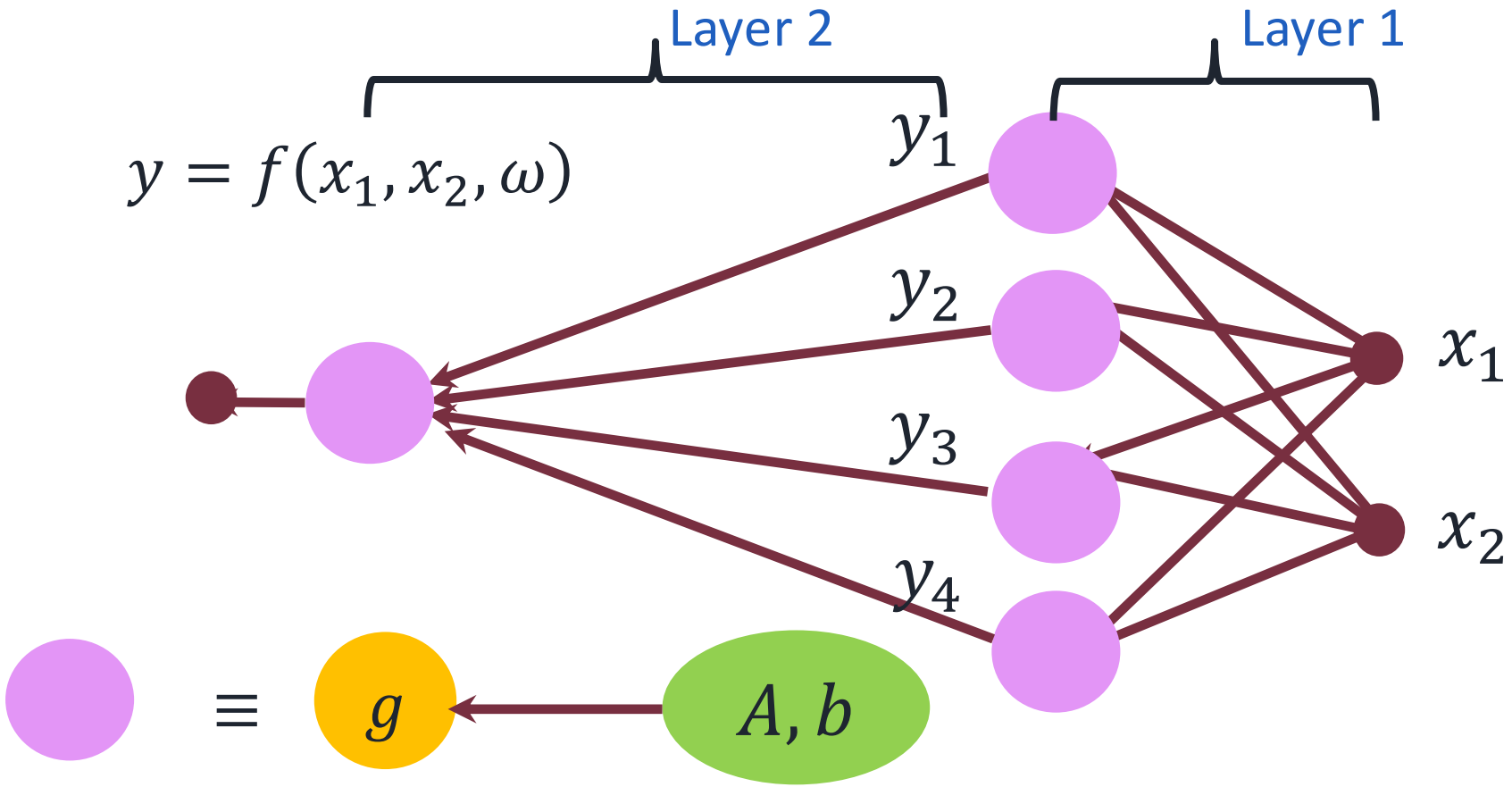
$$x = (-2, 1, 4), \qquad \boldsymbol{A} = \begin{pmatrix} 2 & -3 & 1 \\ 1 & 2 & 3 \end{pmatrix}, \qquad \boldsymbol{b} = (-5, -4)$$



$$y = g\left((-2,1,4)\begin{pmatrix} 2 & -3 & 1 \\ 1 & 2 & 3 \end{pmatrix}^{\mathrm{T}} + (-5,-4)\right)$$

Multi-Node Perceptron

Consider modeling the data triplets $D = \{(x_1, x_2, y)_{i=1}^{N}\}$ with a function of the form $y = f(x_1, x_2, \omega)$.



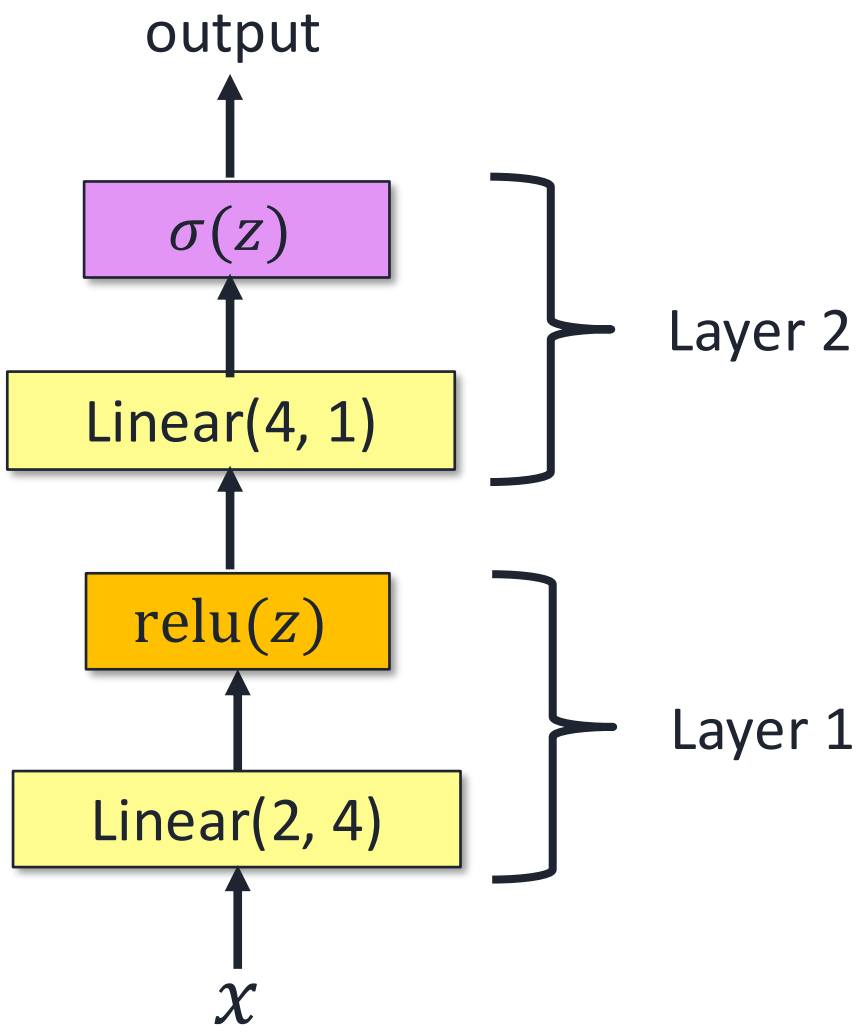
Multi-Node Perceptron

Given the complexity of ML models (i.e., ML functions), it is now common to use a graphical representation that uses higher-level components. For example,

$$f(z) = zA^T + b$$

$$g(z) = \text{relu}(z)$$

$$\sigma(z) = (1 + e^{-z})^{-1}$$

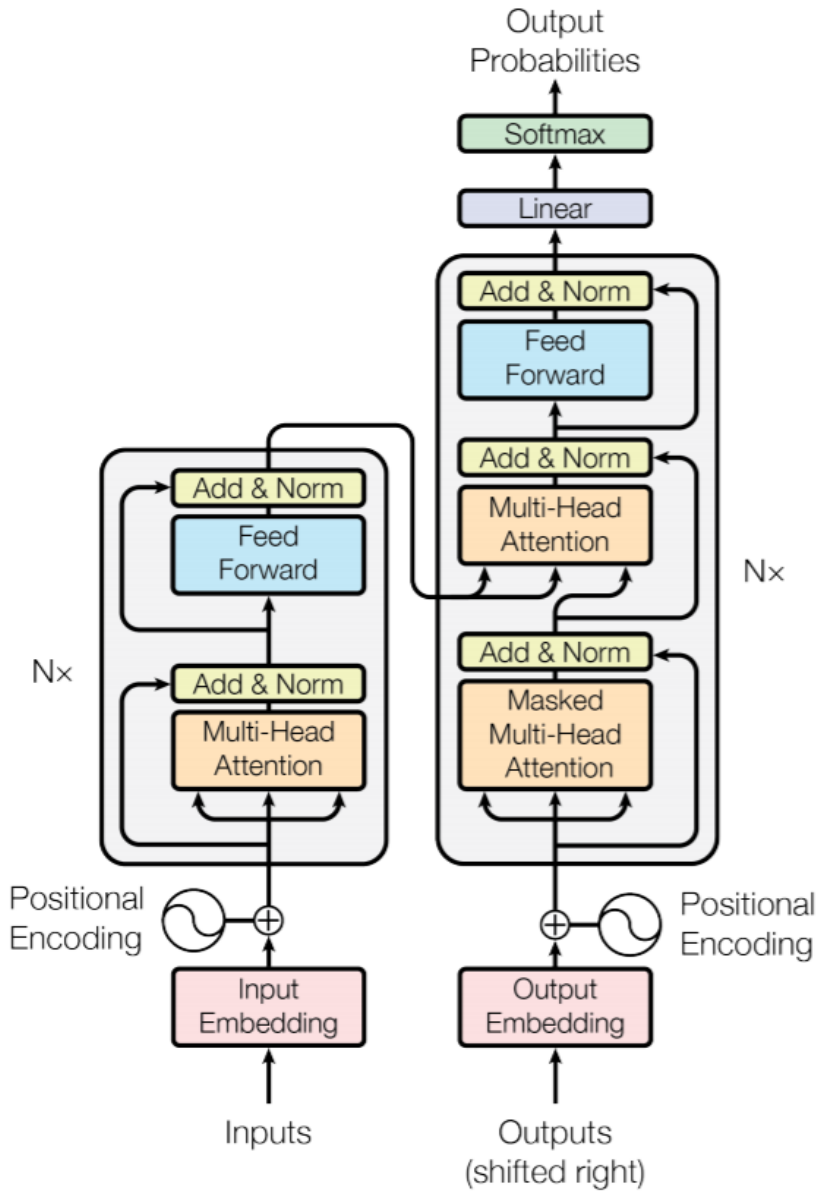


Large Language Model

Here, for example, is high-level graphical representation of an ML model called a **transformer**.

In ChatGPT 3, there are 96 layers with the same architecture, an instance of which is depicted as the two vertical grey boxes.

(Notice, the color-coding differs from ours!)



Summary

- **General Approaches**
  - Supervised, unsupervised, generative, and reinforcement learning.
- **Deep Learning**
  - Uses functions constructed through deep composition.
- **The Perceptron**
  - Basic computational unit: matrix multiplication and addition followed by a nonlinear map.
  - We described the multi-node perceptron and how machine learning models are represented graphically.